

1. The positive integers N and N^2 both end in the same 3-digit number $\overline{abc}$ where $a \neq 0$. What is the sum of all possible values of $\overline{abc}$?
2. Find the remainder when 7^{676} is divided by 240.
3. Define $d(n)$ as the number of positive integer divisors of n . Find the minimum positive integer k such that

$$d(16000k) = d(6400k).$$

4. Find the least odd prime factor of $2019^8 + 1$
5. Find the last two digits of

$$\left\lfloor \frac{3^{66}}{67} \right\rfloor$$

6. Find the sum of all values of positive integers $n \leq 42$ such that 42 divides

$$\sum_{i=0}^{41} n^i.$$

7. Let r be the remainder when $2017^{2025!} - 1$ is divided by $2025!$. Compute $\frac{r}{2025!}$.
(Note that 2017 is prime)
8. Define $\varphi(n)$ as the number of positive integers $k \leq n$ such that $\gcd(k, n) = 1$. Find the amount of values of positive integers $m \leq 42$ such that the amount of positive integers n in which

$$m \mid \sum_{i=0}^{m-1} n^i$$

is equal to $\varphi(m)$.